

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSEAI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)

Architectural Modelling (CO2)

Group No : 3

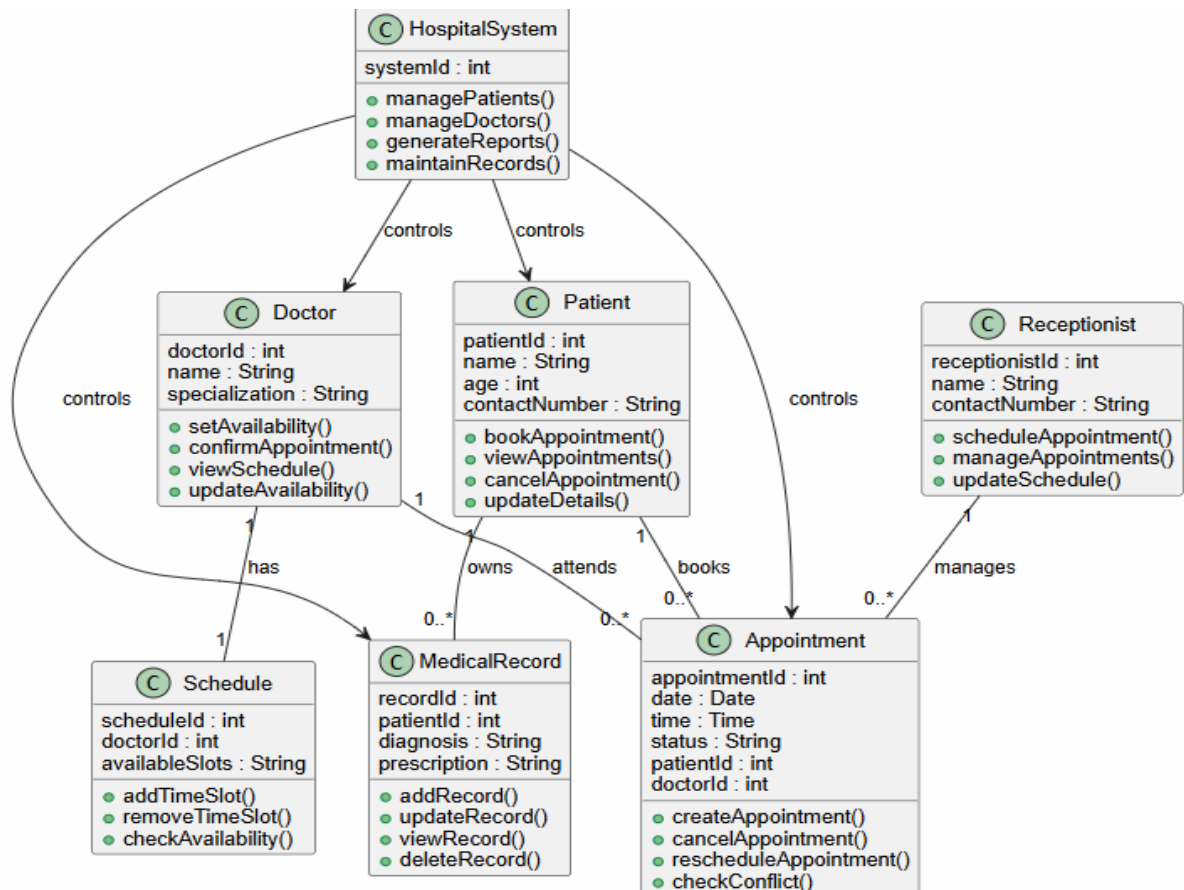
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1. Problem Title : Hospital Appointment Management System

STRUCTURAL DESIGN

2. Class diagram



BEHAVIUIORAL DESIGN

3. Use-case diagram

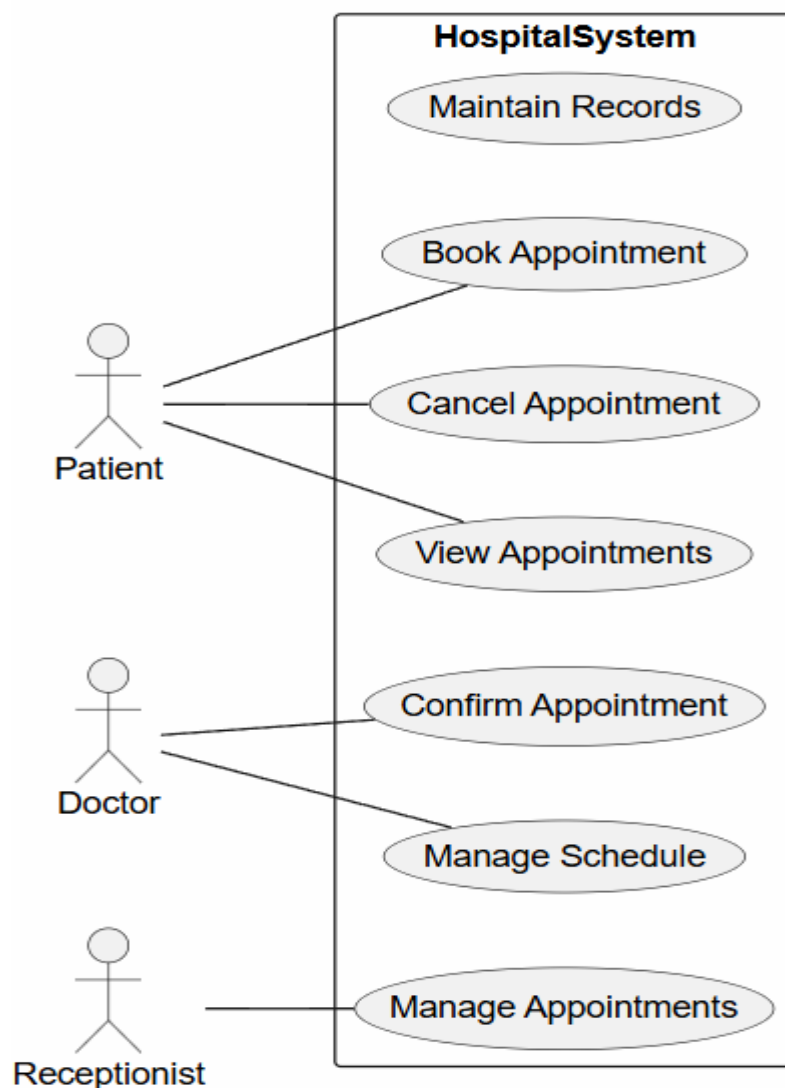
a. List of actors

1. Patient
2. Doctor
3. Receptionist

b. List of use-cases

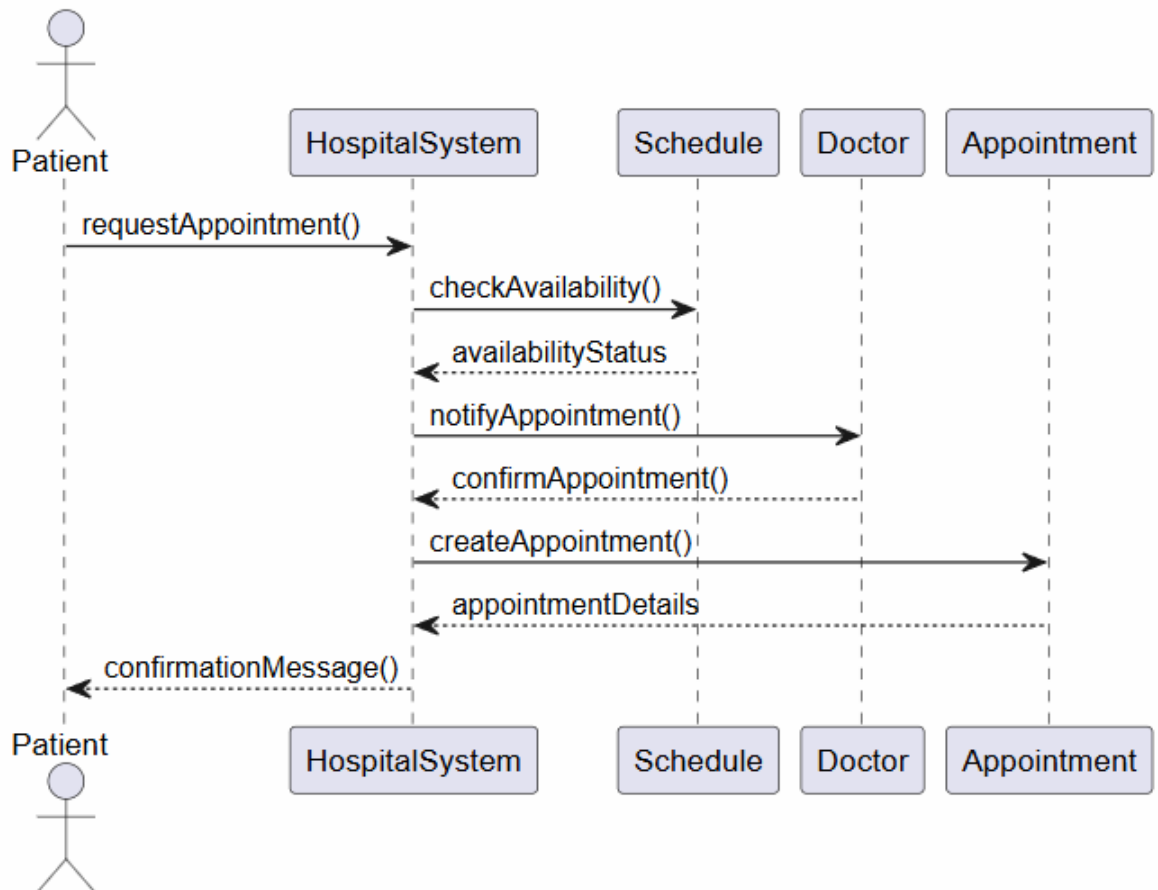
1. **Book Appointment** – Patient selects a doctor and books an available time slot.
2. **View Appointments** – Patient checks scheduled appointments.
3. **Cancel Appointment** – Patient cancels an existing booking.
4. **Manage Schedule** – Doctor updates availability and time slots.
5. **Confirm Appointment** – Doctor approves scheduled appointments.
6. **Manage Appointments** – Receptionist organizes and updates bookings.
7. **Maintain Records** – System stores patient medical history digitally.

c. Use case diagram



4. Sequence diagram (use UML notations)
- a. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

A patient requests an appointment. The system checks doctor availability, confirms with the doctor, and creates the appointment.



- b. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

Receptionist schedules an appointment by checking availability and confirming booking.

